

Networks and Security

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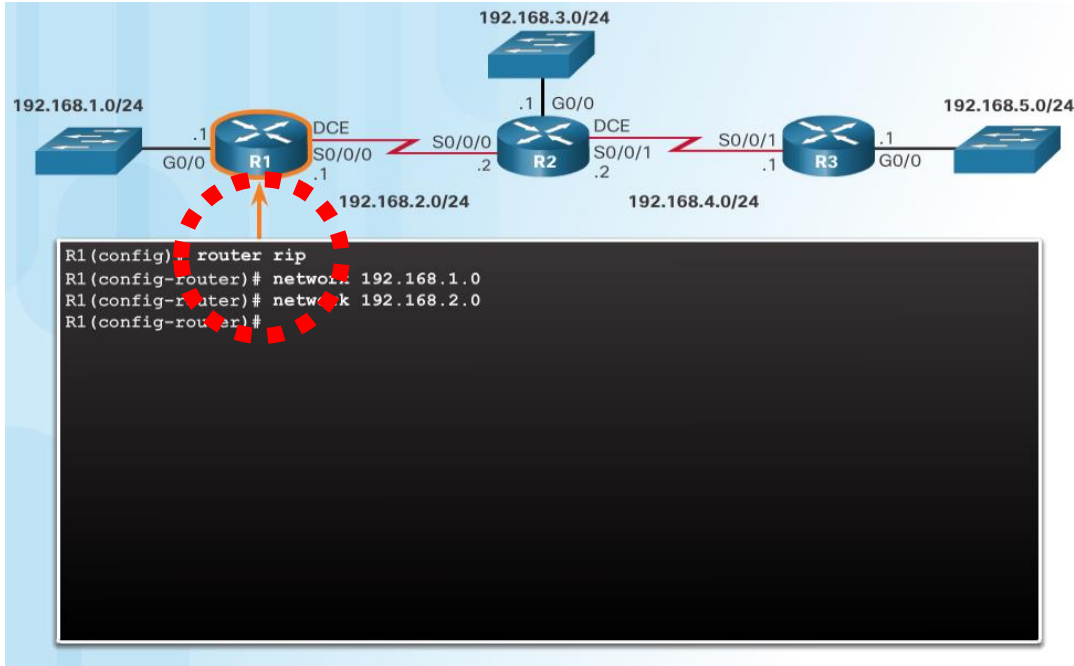
Routing Information Protocol (RIP)

What is Routing Information Protocol (RIP)?

- Routing Information Protocol (RIP) is a distance vector protocol that uses hop count as its primary metric.
- RIP, which was defined in RFC 1058 in 1988, is known for being easy to configure and easy to use in small networks.
- RIP is a distance-vector routing protocol, which employs the hop count as a routing metric.
- RIP prevents routing loops by implementing a limit on the number of hops allowed in a path from the source to a destination.
- The maximum number of hops allowed for RIP is 15.
- Originally each RIP router transmitted full updates every 30 seconds.

Enabling RIP on a Cisco router

- The **network** *network-address* router configuration mode command:
 - Enables RIP on all interfaces that belong to a specific network
 - Advertises the network in RIP routing updates sent to other routers every 30 seconds.



RIP configuration

Router# show ip rip database

Displays all the routes known by RIP—even routes that are not currently in the routing table

Router# show ip protocols

Displays detailed information about dynamic routing processes currently running on the router

```
R1# show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 16 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip

  Default version control: send version 1, receive any version
    Interface          Send  Recv  Triggered RIP  Key-chain
  GigabitEthernet0/0    1     1 2
  Serial0/0/0          1     1 2

  Automatic network summarization is in effect
  Maximum path: 4
  Routing for Networks:
    192.168.1.0
    192.168.2.0

  Routing Information Sources:
    Gateway         Distance    Last Update
    192.168.2.2      120        00:00:15
  Distance: (default is 120)

R1#
```

RIP configuration (cotd.)

Router# show ip route 172.16.3.0

Displays detailed information about the route 172.16.3.0

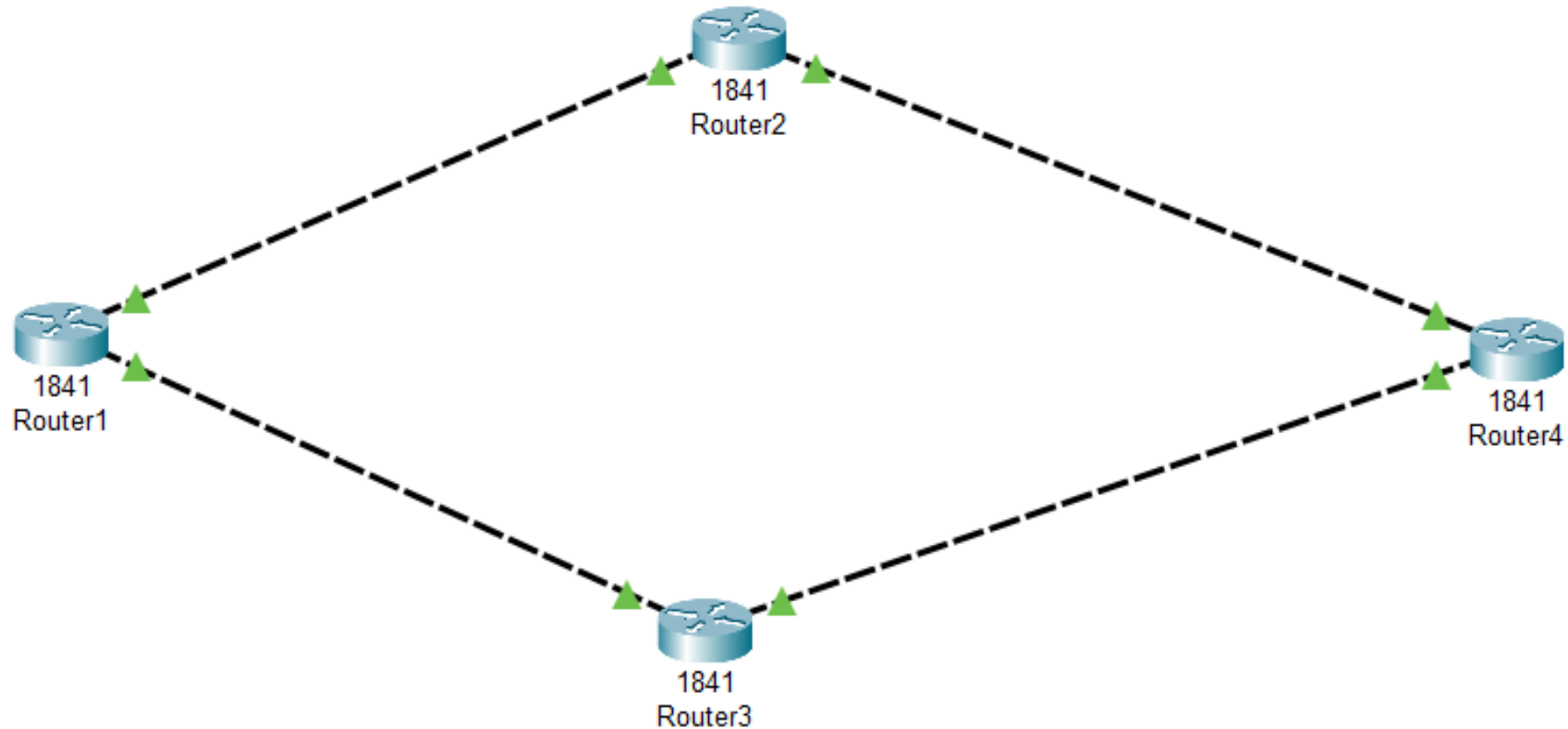
```
R1# show ip route | begin Gateway
Gateway of last resort is not set

      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected,
GigabitEthernet0/0
L       192.168.1.1/32 is directly connected,
GigabitEthernet0/0
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, Serial0/0/0
L       192.168.2.1/32 is directly connected, Serial0/0/0
R       192.168.3.0/24 [120/1] via 192.168.2.2, 00:00:24,
Serial0/0/0
R       192.168.4.0/24 [120/1] via 192.168.2.2, 00:00:24,
Serial0/0/0
R       192.168.5.0/24 [120/2] via 192.168.2.2, 00:00:24,
Serial0/0/0
R1#
```

Exercise

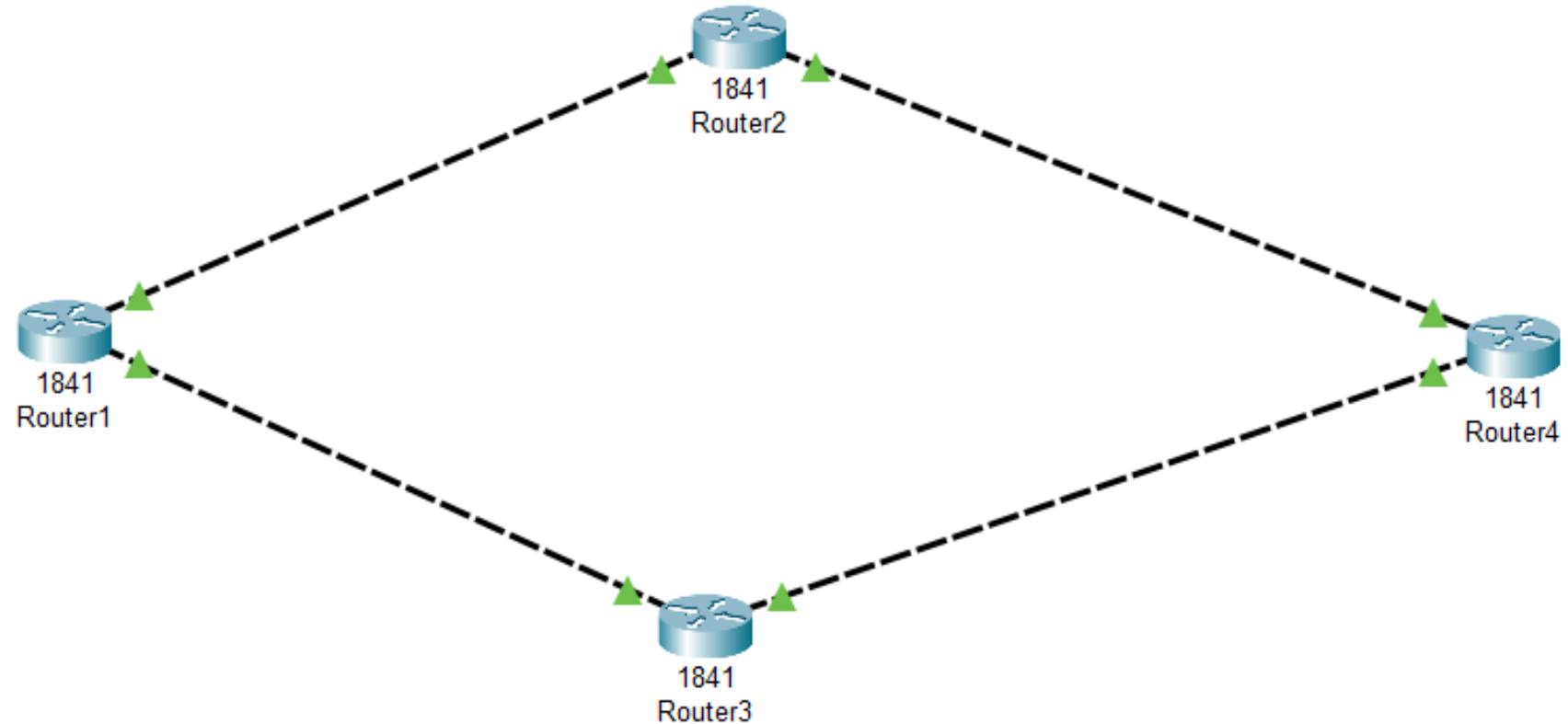
Objectives

Design the given topology and configure RIP to demonstrate that all routers are successfully connected.



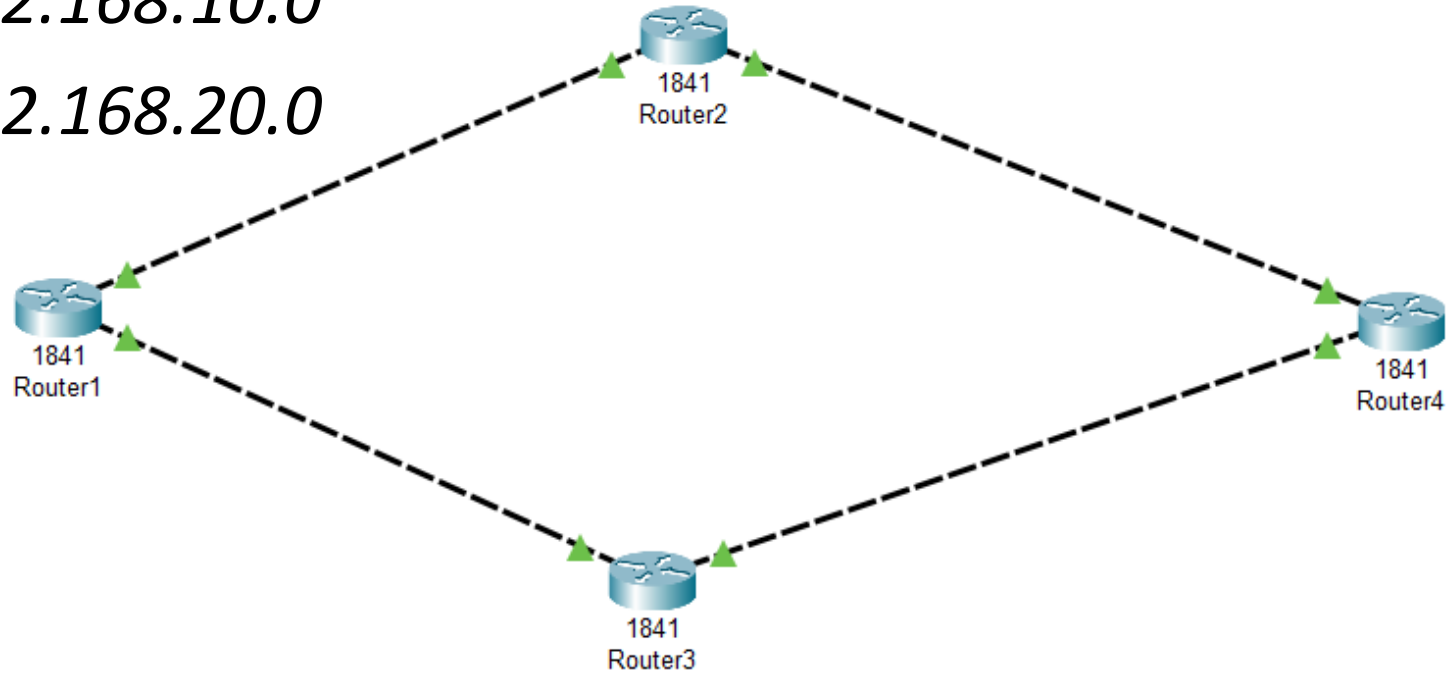
RIP configuration

- R1:
192.168.10.1
192.168.20.1
- R2:
192.168.20.2
192.168.30.1
- R3:
192.168.10.2
192.168.40.1
- R4:
192.168.30.2
192.168.40.2



Configuring RIP with the CLI

- *R1(config)#router rip*
- *R1(config-router)#network 192.168.10.0*
- *R1(config-router)#network 192.168.20.0*
- *Continue with other routers*



Check the status

- Are you able to ping from each router to other ones?

